

6(a) Work function is the minimum amount of energy required for an electron to escape from the surface of a metal. B1

6(b)(i) Using conservation of energy,

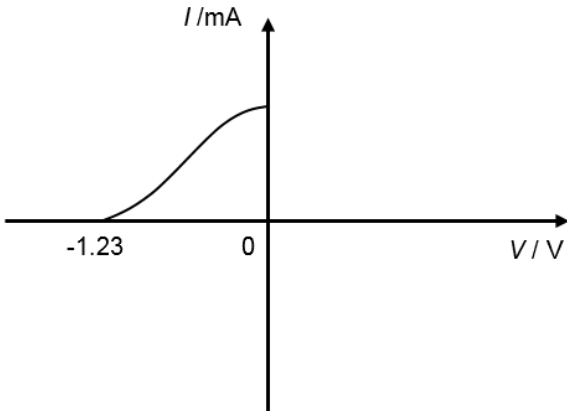
$$KE_{\max} = hf - \phi \quad \text{M1}$$

$$= \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{4.0 \times 10^{-7}} - 3.0 \times 10^{-19}$$

$$= 1.97 \times 10^{-19} \text{ J} \quad \text{C1}$$

$$= \frac{1.97 \times 10^{-19}}{1.60 \times 10^{-19}} = 1.23 \text{ eV} \quad \text{A1}$$

6(b)(ii) Some of the electrons in material C may need energy greater than the work function energy to be liberated as photoelectrons as they are beneath the metal surface. Hence, not all photoelectrons have this maximum kinetic energy. B1

Question	Marks
6(b)(iii) p.d. = 1.23 V	B1
6(c)	
	
Correct shape of curve (Only in 1 quadrant of graph) Stopping potential clearly and correctly labelled numerically.	B1 B1
6(d) When the polarity is being reversed, <u>C will always be at a lower potential compared to D</u> . Photoelectrons released from C will therefore <u>always be attracted to D</u> and close the circuit for current to flow through the milliammeter.	M1 A1
6(e) Metal surface is oxidized. OR	B1
The intensity of the radiation is not sufficiently high enough to liberate enough photoelectrons for them to be measureable by the milliammeter.	
Any one of the above points.	